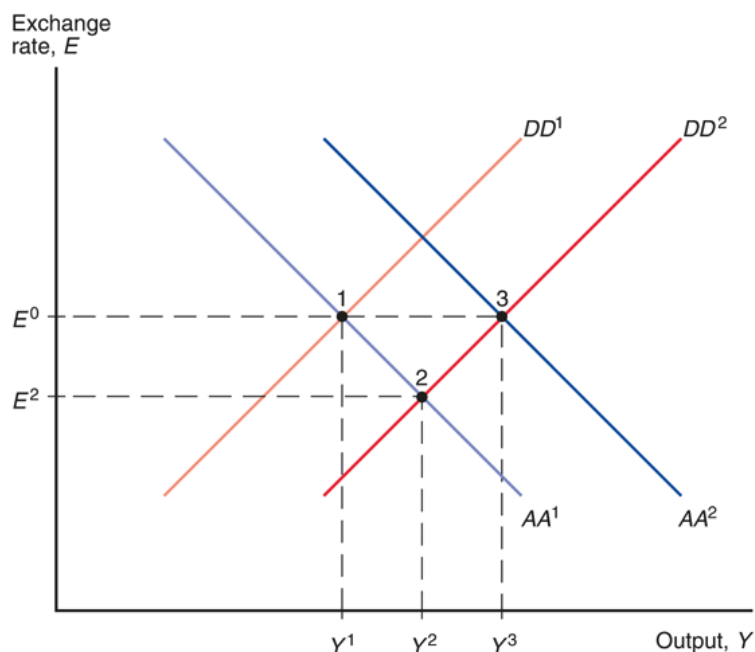


ECON 1550 Spring 2026: Problem Set 7 Answer Key

1. (a) A temporary tax cut raises disposable income and consumption, so the DD curve shifts to the right. A temporary increase in the money supply shifts the AA curve to the right. Because both policies are expansionary, output rises unambiguously.

The initial equilibrium is point 1. The tax cut alone shifts the DD curve from DD^1 to DD^2 , with the equilibrium moving from point 1 to point 2. Adding the temporary monetary expansion shifts the AA curve from AA^1 to AA^2 , moving the equilibrium from point 2 to point 3.



In the figure, the shifts in AA and DD are such that the exchange rate at point 1 is the same as the exchange rate at point 3. In general, however, the effect on the exchange rate is ambiguous. If the DD shift is relatively large, the new equilibrium can have a lower exchange rate than E^0 . If the AA shift is relatively large, the new equilibrium can have a higher exchange rate than E^0 .

Summing up, the combined temporary tax cut and temporary monetary expansion raise output, while the effect on the exchange rate is ambiguous.

(b) The expansionary money supply announcement causes the expected exchange rate to go up and shifts the AA curve to the right. This leads to an immediate increase in output and an immediate currency depreciation. The effects of the anticipated policy action thus precede the policy's actual implementation.

(c) A “buy American” provision results in a larger permanent rightward shift in the DD curve than an unconstrained increase in government spending because there is a greater demand for U.S. goods than if some imported goods are purchased with the stimulus funds.

In both cases, the short-run expected exchange rate declines, shifting the AA curve down so that it intersects the DD curve at the point where output equals its full-employment level, that is, $Y = Y^f$. Since the short-run equilibrium has $Y = Y^f$, the AA and DD curves do not shift after the short run, so the long-run equilibrium coincides with the short-run equilibrium.

We conclude that the buy American provision and the unconstrained government spending case result in the same $Y = Y^f$ level of output in the short run and in the long run.

Because the buy American provision shifts the DD more than the unconstrained spending, the shift in the AA required to have $Y = Y^f$ is also larger for the buy American provision, and the currency appreciates more.

Additional information (not part of the answer): Because the buy American provision shifts the DD curve farther to the right than unconstrained government spending, keeping equilibrium at $Y = Y^f$ requires a larger downward shift in the AA curve. As a result, the domestic currency appreciates more under the buy American provision than under unconstrained government spending.

(d) When spending is temporary, a “buy American” provision has a larger effect on output in the short run, since the AA curve does not shift because exchange rate expectations remain unchanged.

In the long run, the economy returns to $Y = Y^f$ in both cases, making the long-run effect on output the same with and without “buy American” provisions.

2. (a) Write imports as

$$IM(q, Y - T) = qV(q, Y - T).$$

The price effect is that when q rises, each imported unit costs more in terms of domestic goods, which tends to raise IM for a given import volume. The volume effect is that when q rises, domestic residents buy fewer imported units, which tends to lower IM .

In this model,

$$V(q, Y - T) = \frac{5 + 2(Y - T)}{20q} - \frac{1}{10}.$$

Holding $Y - T$ fixed, a rise in q lowers the term $\frac{5+2(Y-T)}{20q}$, so import volume is decreasing in q .

Imports (expressed in units of domestic goods) are:

$$IM(q, Y - T) = q \left(\frac{5 + 2(Y - T)}{20q} - \frac{1}{10} \right) = \frac{5 + 2(Y - T)}{20} - \frac{q}{10}.$$

Since IM is decreasing in q , the fall in import volume more than offsets the higher price per imported unit. In this model, the volume effect dominates the price effect.

(b) From part (a) and setting $T = 0$,

$$IM = \frac{5 + 2Y}{20} - \frac{q}{10} = \frac{1}{4} + \frac{Y}{10} - \frac{q}{10}.$$

Using $Y^* = 1$ and the equation for exports given by the problem,

$$EX = \frac{1}{4} + \frac{q}{10}.$$

The current account is

$$CA = EX - IM = \left(\frac{1}{4} + \frac{q}{10} \right) - \left(\frac{1}{4} + \frac{Y}{10} - \frac{q}{10} \right) = \frac{q}{5} - \frac{Y}{10}.$$

Aggregate demand for domestic goods is

$$D = C + I + G + CA = \frac{1}{2}Y + \frac{1}{5} + \frac{1}{5} + \frac{q}{5} - \frac{Y}{10} = \frac{2}{5}Y + \frac{2}{5} + \frac{q}{5}.$$

Goods-market equilibrium requires $Y = D$, so

$$Y = \frac{2}{5}Y + \frac{2}{5} + \frac{q}{5}.$$

Re-arranging,

$$\frac{3}{5}Y = \frac{2}{5} + \frac{q}{5},$$

so

$$q = 3Y - 2.$$

Because $q = EP^*/P$ and $P^* = 1$, the DD curve is

$$E = P(3Y - 2) = 3PY - 2P.$$

Points on the DD curve are pairs (E, Y) that are consistent with equilibrium in the market for domestic goods.

In this version of the model, DD is a straight line. Its slope is $3P$, and its vertical intercept is $-2P$.

The DD curve is upward sloping. Take T, I, G, Y^*, P and P^* as given (fixed). A nominal depreciation (E increases) leads to a real depreciation (q increases). Exports EX increase, imports IM decrease, and the current account CA improves (increases). Therefore, demand for the domestic good D increases. In equilibrium, output Y must increase so that the supply matches the increased demand of domestic goods. Higher output also means higher income, which increases demand, which increases output, which increases income, and so on.

(c) Using $M^s = 1$ and the equation for real money demand given by the problem, the equilibrium condition in the money market is

$$\frac{1}{P} = \frac{Y}{1 + R}.$$

So

$$1 + R = PY,$$

and therefore

$$R = PY - 1.$$

Uncovered interest parity is

$$R = \frac{E^e}{E} - 1,$$

because $R^* = 0$.

Plugging the UIP R into the equilibrium condition for the money market gives:

$$\frac{E^e}{E} - 1 = PY - 1.$$

Solving for E gives the AA curve:

$$E = \frac{E^e}{PY}.$$

Points on the AA curve are pairs (E, Y) that are consistent with equilibrium in the money market together with equilibrium in the foreign-exchange market.

The AA curve is downward sloping. Take P, E^e, M^s , and R^* as given (fixed). When real income Y increases, real money demand goes up. Real money supply does not change, since M^s and P are fixed. Therefore, the interest rate R must increase to make bonds more attractive relative to money, and exactly offset the increase in the real money demand caused by the higher Y . In the foreign exchange market, equilibrium requires that the

dollar return on the foreign bond increases to match the increase in the domestic interest rate R . Since E^e and R^* are fixed, the increase in the dollar return on the foreign bond must be achieved by a lower E , that is, by an appreciation.

(d) The initial long-run equilibrium is

$$Y_0 = 1, \quad E_0 = 1, \quad q_0 = 1, \quad R_0 = 0, \quad P_0 = 1, \quad \pi_0 = 0.$$

We now show how to find these values.

With the given parameter values, the Phillips curve becomes

$$\pi = Y - 1.$$

In the long run, output equals the full-employment level of output:

$$Y_0 = Y^f = 1.$$

So inflation is

$$\pi_0 = Y_0 - 1 = 0.$$

Alternatively, we can deduce π_0 in the long run from the assumption that $\pi^e = 0$ and the long-run condition $\pi^e = \pi$.

In the long run, expected and actual exchange rates are equal:

$$E_0^e = E_0.$$

UIP and $R^* = 0$ then give

$$R = R^* + \frac{E_0^e}{E_0} - 1 = R^* = 0.$$

The AA curve from part (c) in the initial equilibrium is

$$E_0 = \frac{E_0^e}{P_0 Y_0}.$$

Using $E_0^e = E_0$ and $Y_0 = Y^f = 1$ gives

$$P_0 = 1.$$

The DD relation from part (b) is

$$q = 3Y - 2.$$

At $Y_0 = 1$,

$$q_0 = 1.$$

Since $q_0 = E_0/P_0$ and $P_0 = 1$,

$$E_0 = E_0^e = 1.$$

The remaining endogenous variables are

$$Y_0 - T = 1, \quad C_0 = \frac{1}{2}, \quad EX_0 = \frac{7}{20},$$

$$IM_0 = \frac{1}{4}, \quad CA_0 = \frac{1}{10}, \quad D_0 = 1.$$

(e) With the tariff,

$$V((1 + \tau)q, Y) = \frac{5 + 2Y}{20(1 + \tau)q} - \frac{1}{10},$$

because $T = 0$.

Imports are therefore

$$IM(q, \tau, Y) = qV((1 + \tau)q, Y) = q \left(\frac{5 + 2Y}{20(1 + \tau)q} - \frac{1}{10} \right) = \frac{5 + 2Y}{20(1 + \tau)} - \frac{q}{10}.$$

So the current account is

$$CA(q, \tau, Y) = \left(\frac{1}{4} + \frac{q}{10} \right) - \left(\frac{5 + 2Y}{20(1 + \tau)} - \frac{q}{10} \right).$$

A higher tariff makes the term $\frac{5+2Y}{20(1+\tau)}$ smaller. So imports fall and the current account rises.

The intuition is that at a fixed real exchange rate and a fixed level of output, the tariff makes imported goods more expensive for domestic buyers. Domestic residents therefore buy fewer imports, and that improves the current account.

(f) From part (e),

$$CA(q, \tau, Y) = \left(\frac{1}{4} + \frac{q}{10} \right) - \left(\frac{5 + 2Y}{20(1 + \tau)} - \frac{q}{10} \right).$$

Re-arranging,

$$CA = \frac{1}{4} - \frac{1}{4(1 + \tau)} + \frac{q}{5} - \frac{Y}{10(1 + \tau)}.$$

Aggregate demand for domestic goods is

$$D = \frac{1}{2}Y + \frac{1}{5} + \frac{1}{5} + \frac{1}{4} - \frac{1}{4(1+\tau)} + \frac{q}{5} - \frac{Y}{10(1+\tau)}.$$

So

$$D = \frac{1}{2}Y + \frac{13}{20} - \frac{1}{4(1+\tau)} + \frac{q}{5} - \frac{Y}{10(1+\tau)}.$$

The equilibrium condition in the market for domestic goods $Y = D$ gives

$$Y = \frac{1}{2}Y + \frac{13}{20} - \frac{1}{4(1+\tau)} + \frac{q}{5} - \frac{Y}{10(1+\tau)}.$$

Move the Y -terms to the left:

$$\left(\frac{1}{2} + \frac{1}{10(1+\tau)}\right)Y = \frac{13}{20} - \frac{1}{4(1+\tau)} + \frac{q}{5}.$$

So

$$\frac{6+5\tau}{10(1+\tau)}Y = \frac{8+13\tau}{20(1+\tau)} + \frac{q}{5}.$$

Solving for q ,

$$q = \frac{(12+10\tau)Y - 8 - 13\tau}{4(1+\tau)}.$$

Because $q = EP^*/P$ and $P^* = 1$, the DD curve is

$$E = P \left[\frac{12+10\tau}{4(1+\tau)}Y - \frac{8+13\tau}{4(1+\tau)} \right].$$

This is again a straight line with slope

$$m_{DD}(\tau) = P \frac{12+10\tau}{4(1+\tau)} = P \frac{6+5\tau}{2(1+\tau)},$$

and intercept

$$b_{DD}(\tau) = -P \frac{8+13\tau}{4(1+\tau)}.$$

Without tariffs, the slope is $3P$ and the intercept is $-2P$. With a positive tariff, the slope is smaller because

$$\frac{6+5\tau}{2(1+\tau)} < 3,$$

and the intercept is lower because

$$\frac{8 + 13\tau}{4(1 + \tau)} > 2.$$

So in an E -against- Y graph, the tariff DD curve is flatter and lies below the no-tariff DD curve. That is the same as saying DD shifts to the right.

The intuition is as follows. For a given E , we found in part (e) that a tariff reduces imports. Domestic demand is redirected away from foreign goods and toward domestic goods. In equilibrium, output goes up to meet the increased demand for domestic goods.

(g) The tariff does not enter the money market or uncovered interest parity. So the AA curve is the same as before:

$$E = \frac{E^e}{PY}.$$

AA is not a straight line. So it does not have one constant slope or a finite vertical intercept.

If the tariff is temporary, it leaves E^e unchanged, so AA is unchanged. If the tariff is permanent, the DD shifts to the right permanently, inducing a permanent appreciation. Therefore, the long-run exchange rate goes down. In the short run, the expected exchange rate goes down as well, since it equals the long-run exchange rate. With a lower E^e , the AA gives a lower E for any given Y and P , which means that the AA curve shifts down.

The intuition is that the AA curve comes from equilibrium in money and FX markets. Current tariffs work through the goods market, not directly through money demand or interest parity. AA moves only if the tariff changes the expected future exchange rate. The expected future exchange rate only changes if the actual long-run exchange rate changes. When the tariff is temporary, the long-run does not change, so the AA does not shift. When tariffs are permanent, the permanently lower demand for foreign goods causes a permanent appreciation.

(h) Since the shock is transitory, the long-run equilibrium after the temporary tariff shock is the initial long-run equilibrium found in part (d):

$$Y_{LR} = 1, \quad E_{LR} = 1, \quad q_{LR} = 1, \quad P_{LR} = 1.$$

Therefore, in the short run,

$$E_{SR}^e = E_{LR} = 1.$$

The short-run price level is fixed at the initial value:

$$P_{SR} = P_0 = 1.$$

The short run AA is unchanged because the shock is temporary so E^e is unchanged:

$$AA_{SR} : \quad E = \frac{1}{Y}.$$

The DD curve shifts to the right. With $\tau = 1/3$ and $P = 1$, the tariff DD curve is

$$DD_{SR} : \quad E = \frac{46Y - 37}{16}.$$

The short-run equilibrium solves

$$\frac{46Y_{SR} - 37}{16} = \frac{1}{Y_{SR}},$$

so

$$46Y_{SR}^2 - 37Y_{SR} - 16 = 0.$$

The positive solution is

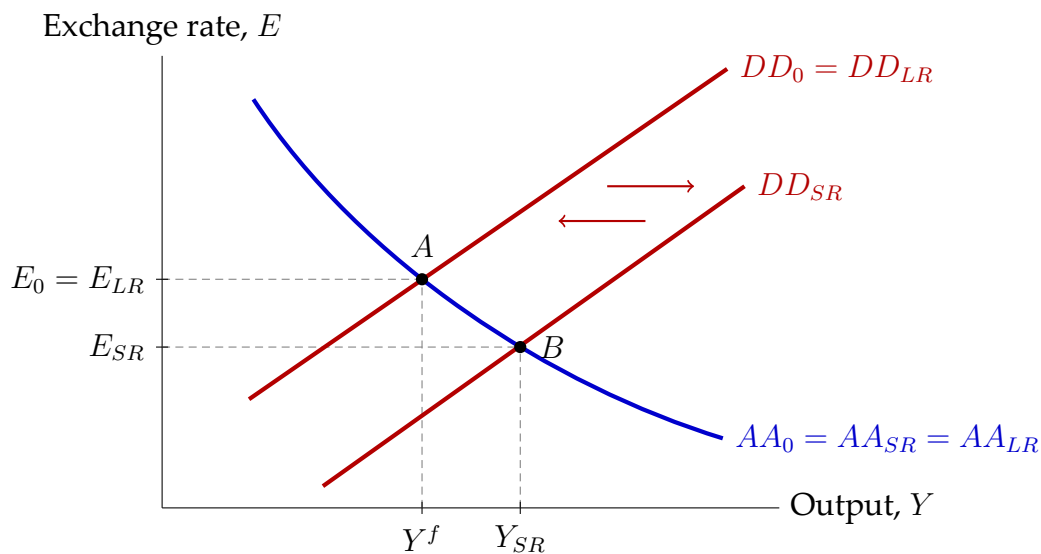
$$Y_{SR} = 1.116.$$

Therefore

$$E_{SR} = \frac{1}{Y_{SR}} = 0.896, \quad q_{SR} = \frac{E_{SR}}{P_{SR}} = 0.896.$$

From the initial equilibrium to the short-run equilibrium, the economy moves from the initial point A to the short-run point B , with higher Y and lower E . The tariff shifts spending toward domestic goods. That raises output and the exchange rate appreciates.

Immediately after the short run, the tariff returns to zero. The DD shifts back to its original position (the left arrow in the figure), and the economy returns from B to A , the initial long-run equilibrium, where it stays forever after. The right arrow shows the initial rightward shift of DD when the tariff is imposed.



(i) At the initial long-run equilibrium from part (d),

$$E_0 = 1, \quad q_0 = 1, \quad Y_0 = 1, \quad P_0 = 1, \\ \pi_0 = 0, \quad CA_0 = \frac{1}{10}, \quad EX_0 = \frac{7}{20}, \quad IM_0 = \frac{1}{4}.$$

At the short-run equilibrium from part (h),

$$Y_{SR} = 1.116, \quad E_{SR} = 0.896, \quad q_{SR} = 0.896, \quad P_{SR} = 1, \quad \pi_{SR} = 0.$$

$$EX_{SR} = \frac{1}{4} + \frac{1}{10}q_{SR} = 0.340,$$

$$IM_{SR} = \frac{5 + 2Y_{SR}}{20(1 + 1/3)} - \frac{q_{SR}}{10} = 0.182,$$

$$CA_{SR} = EX_{SR} - IM_{SR} = 0.158.$$

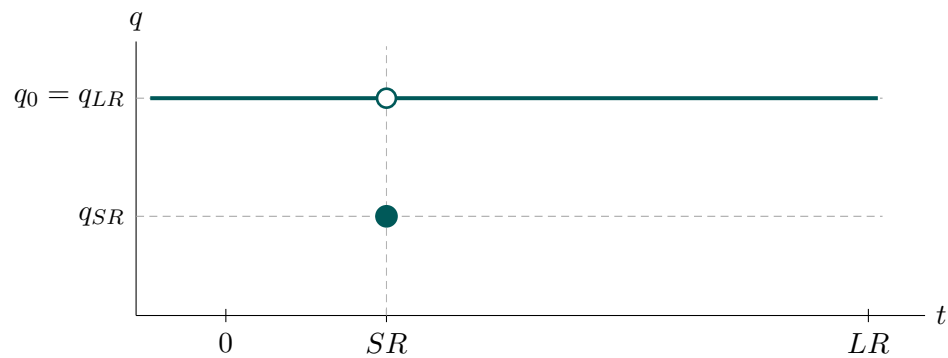
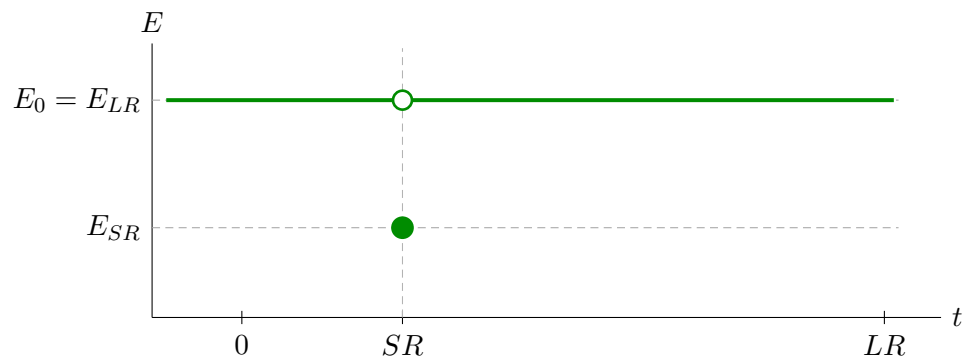
After the short run, the economy returns to the initial long-run equilibrium.

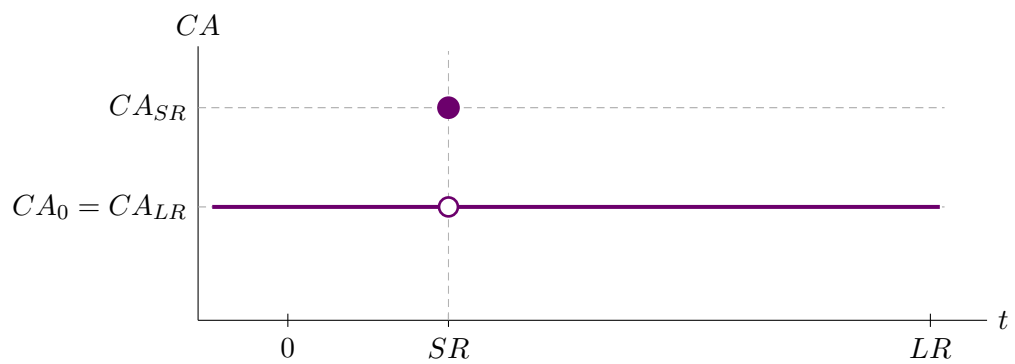
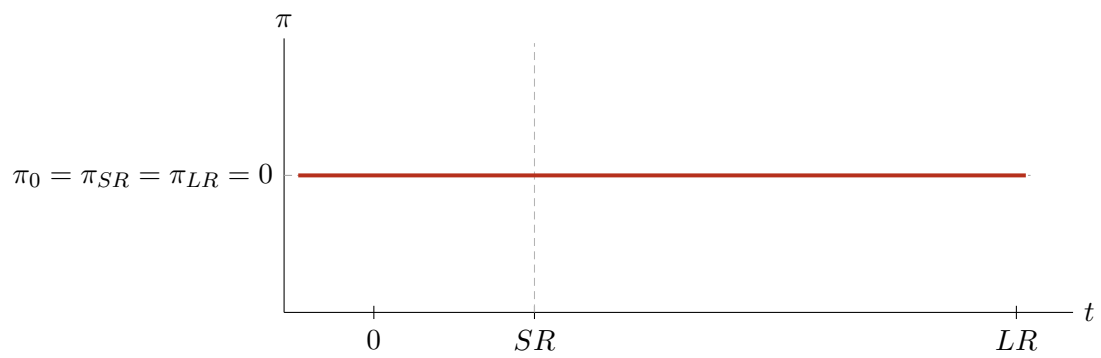
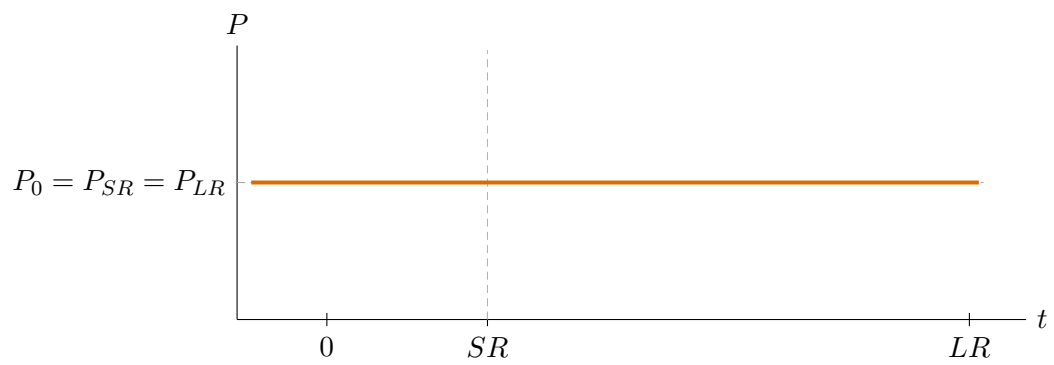
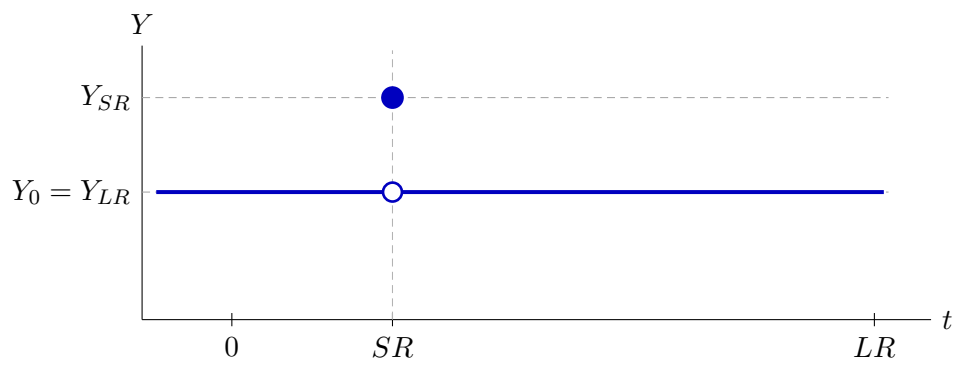
A summary table for the values at 0, SR, and LR is

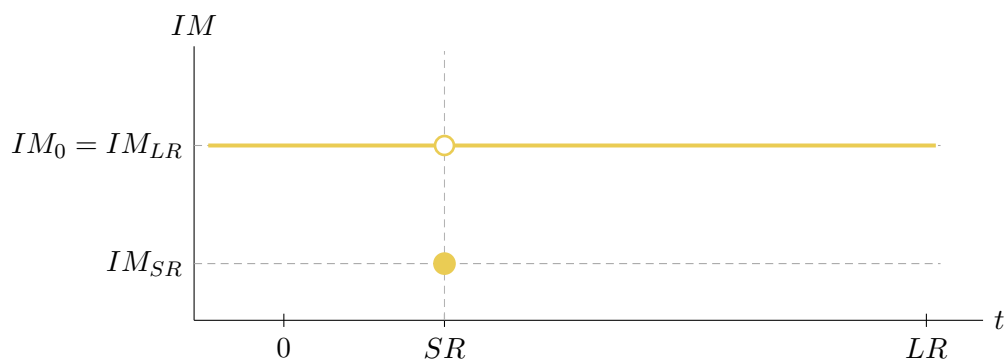
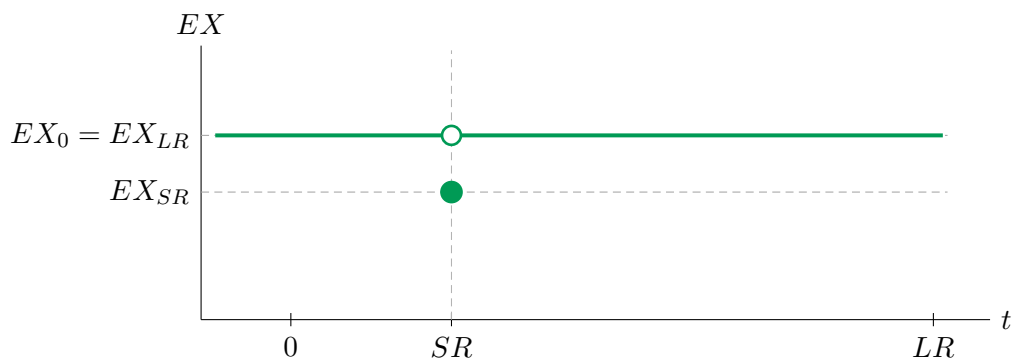
Variable	0	SR	LR
E	1.000	0.896	1.000
q	1.000	0.896	1.000
Y	1.000	1.116	1.000
P	1.000	1.000	1.000
π	0.000	0.000	0.000
CA	0.100	0.158	0.100
EX	0.350	0.340	0.350
IM	0.250	0.182	0.250

Because the temporary tariff is treated as a short-run impact only, each variable jumps to its impact value at SR and then jumps back to the initial equilibrium's value immediately after. So E and q jump down and then rise back, Y jumps up and then falls back, P and π are unchanged throughout, CA jumps up and then falls back, and both EX and IM jump down and then rise back to their original values.

One set of qualitatively correct paths is shown below.







(j) Now the tariff stays at $1/3$ in both the short run and the long run:

$$\tau_{SR} = \tau_{LR} = \frac{1}{3}.$$

In the long run, output is still at full employment:

$$Y_{LR} = 1.$$

The money supply, foreign interest rate, and full-employment output are unchanged, so the long-run money market still gives

$$P_{LR} = 1, \quad R_{LR} = 0.$$

Use the tariff DD relation from part (f):

$$q = \frac{(12 + 10\tau)Y - 8 - 13\tau}{4(1 + \tau)}.$$

At $Y = 1$ and $\tau = 1/3$,

$$q_{LR} = \frac{46 - 37}{16} = \frac{9}{16}.$$

Since $P_{LR} = 1$, the long-run exchange rate is

$$E_{LR} = q_{LR} = \frac{9}{16}.$$

So on impact,

$$E_{SR}^e = E_{LR} = \frac{9}{16}.$$

The short-run price level is still fixed at

$$P_{SR} = P_0 = 1.$$

With $\tau = 1/3$, the DD curve on impact is

$$DD_{SR} : E = \frac{46Y - 37}{16}.$$

The AA curve on impact is

$$AA_{SR} : E = \frac{9}{16Y}.$$

Solve for the short-run equilibrium:

$$\frac{46Y_{SR} - 37}{16} = \frac{9}{16Y_{SR}}.$$

So

$$46Y_{SR}^2 - 37Y_{SR} - 9 = 0.$$

This factors as

$$(Y_{SR} - 1)(46Y_{SR} + 9) = 0.$$

The positive solution is

$$Y_{SR} = 1.$$

Therefore

$$E_{SR} = \frac{9}{16}, \quad q_{SR} = \frac{9}{16}, \quad R_{SR} = 0.$$

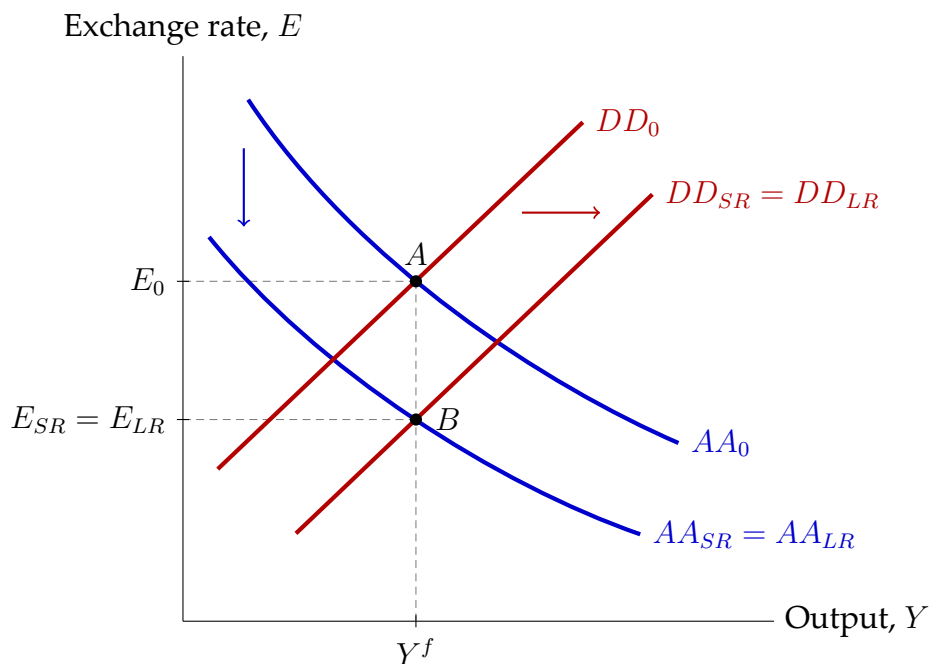
Because $Y_{SR} = Y^f = 1$, inflation is zero:

$$\pi_{SR} = 0.$$

So the price level does not change and the short-run equilibrium is already the long-run equilibrium.

In words, DD shifts to the right (the right arrow in the figure) because the tariff reduces

imports. AA shifts down (the down arrow in the figure) because the permanent tariff lowers the expected future exchange rate. Those two shifts exactly offset each other in terms of output. The economy moves from the initial equilibrium A to the new equilibrium B , which has the same Y as before but a lower E .



(k) From part (j), the short-run and long-run equilibria are the same:

$$Y_{SR} = Y_{LR} = 1, \quad P_{SR} = P_{LR} = 1, \\ E_{SR} = E_{LR} = \frac{9}{16}, \quad q_{SR} = q_{LR} = \frac{9}{16}, \quad \pi_{SR} = \pi_{LR} = 0.$$

Exports and imports are

$$EX_{SR} = EX_{LR} = \frac{1}{4} + \frac{1}{10} \frac{9}{16} = \frac{49}{160} = 0.306, \\ IM_{SR} = IM_{LR} = \frac{5 + 2 \cdot 1}{20(1 + 1/3)} - \frac{1}{10} \frac{9}{16} = \frac{33}{160} = 0.206.$$

So

$$CA_{SR} = CA_{LR} = EX_{SR} - IM_{SR} = \frac{1}{10} = 0.100.$$

The current account does not change because two effects happen to cancel each other in this particular example. The imposition of the tariff reduces imports, but the appreciation increases imports and reduces exports by exactly enough to leave the current account

unchanged. This is not a general result. In general, the CA can deteriorate, stay unchanged, or improve in response to an ad-valorem import tariff (see also [Problem Set 2](#), question 1).

A summary table is

Variable	0	SR	LR
E	1.000	0.5625	0.5625
q	1.000	0.5625	0.5625
Y	1.000	1.000	1.000
P	1.000	1.000	1.000
π	0.000	0.000	0.000
CA	0.100	0.100	0.100
EX	0.350	0.306	0.306
IM	0.250	0.206	0.206

Time paths are shown below. Because the permanent tariff moves the economy directly to the new long-run equilibrium, every path is flat after the impact jump at SR .

